

Beans

CULTIVARS

- Green beans: Jade II, Maxibell
- Yellow beans: Rocdor, Gold Rush

INTENSIVE SPACING

- 2 rows (20" - 50 cm) per bed spaced every 2" (5 cm) along the row. Approximately 1500 seeds/100' bed
- Earthway seeder: bean plate. Seed into 1" (2.5cm) deep furrow

FERTILIZATION

- Like peas, beans will do best in beds amended with compost, fresh or aged. Rule of thumb is to spread 1 inch (5 cm) in the middle of the bed.
- When planting in beds where compost was applied in the previous crops, we only supplement with a mixture of alfalfa meal (2-0-2) and pelleted chicken manure (5-3-2). We add this extra fertility boost at first cultivation 2-3 weeks after the direct sown date.
- Inoculants can increase yields by promoting a rhizobium-bean symbiosis that facilitates nitrogen fixation. If nitrogen-fixing nodules are not found on most legumes grown in a specific planting area (ie new beds), inoculants are added to the seeds when sowing. Follow the manufacturer's instructions.

IRRIGATION

Irrigation Type

- Drip tape only (2 per bed)
- No sprinkler to keep foliage and pods dry and thus prevent diseases

Irrigation Management

- Root depth: Shallow (6"-24")
- General: Can drink lots of water because they are fast growing. Vulnerable to disease with overhead water.

IMPLANTATION

Beans are a popular crop with customers, but have low productivity rates being time-consuming to harvest. To grow this crop profitably, we favour filet-type beans that command better prices and only harvest plants 2 to 3 weeks before moving to a new succession.

For a continuous supply, we simultaneously seed two cultivars with different DTM and make successive sowings every +/- 18 days.

EQUIPMENT

- Rotary harrow
- 30" bed prep rake
- **Direct Seeding:** Earthway seeder + bean plate. 2 tubes for the rake, scale, notepad and your seeds
- **Transplanting:** Popping tray

ITINERARY

- Remove tarp and clean out large debris. Use the rake if needed.*
- Uniformly apply amendments and pass with the rotary harrow at a depth of 1.5" (4 cm). Use the rake to remove debris if needed. Otherwise, the bed is ready to be seeded.
- Mark your 2 rows with the bed prep rake and seed with the Earthway. Write down the total amount of seed dropped onto the first bed and comparing with the target. Adjust calibration if required.
- Install irrigation and water until the soil is deeply moistened. Keep moist for the next 7-10 days until all seeds have germinated. Once the seeds have germinated, favour drip irrigation over overhead if the plants need watering.
- Start some extra seedlings in the greenhouse to fill in the holes on the beds. Ideally, These should be started in the week prior to when the direct seeding is done.
- Minimum soil temperature is 15°C (20°C is optimal). If seeded in soil that is too cold, cover seedbed with clear plastic one week prior to seeding date

*If the previous crop is too well established, or if the bed is too messy, mow the previous crop as low as possible and cover it with soil by raising the bed about an inch using the rotary plow.

WEEDING

- Beans are cultivated using a stirrup hoe +/-14 days after direct sowed and again another 10 days later. if needs be, add fertility to the crop at that time.

PLANT PROTECTION

- Downy mildew is a common condition favoured by the cool, moist conditions in the winter greenhouses. Relying on mildew-resistant varieties is your best bet. Once infected, there is little that can be done.

HARVEST

- Beans are harvested in the fields in a kneeling position using a solid-bottom bins (to prevent dirtying the beans)
- Harvest them when they are 6–7" long (15–18 cm) and as wide as a pencil at the most, and definitely before the seeds become bulbous in the pod.
- To harvest with one hand (so to use both simultaneously), hold the bean between the index finger, middle finger and ring finger, and use the thumb to break the bean off the peduncle.
- Make sure to gather as many beans as you can in one hand before putting them in the bin (to keep long hand motions to a minimum).
- They should be clean and free from any mold or rotting. Harvest every second day to pick them at the right time.
- In as much as possible, harvest when it isn't raining (dry produce keeps better).
- Plan on an average of 2 to 3 hours/100'/row/harvest when the plants are well supplied with beans.

CONDITIONING

- Freshly picked beans should not be washed (they can otherwise form a mold) and must be cooled in the cold room in sealed bins.
- If the beans do get damp, open the bins and let them dry in the cold room for a couple of hours (max 4 hours as they may decrease in quality).

STORING

- Beans are stored in airtight bins and kept cold room kept at 35 °F (2 °C) and should not stay longer than 4 days before being sold.